Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/02/2014
PI Name:	Perera, Imara Y Ph.D.		
Project Title:	The Effect of Spaceflight on Transgenic Arabidopsis Plants With Compromised Signaling		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27695	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	07/01/2010	**End Date:**	06/30/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 ; ISS NOTE: Extended to 6/30/2015 per NSSC (Ed., 5/5/14) NOTE: Extended to 6/30/2014 per A. Chu/ARC (Ed., 5/28/2013)		
Key Personnel Changes/Previous PI:	Caroline M Smith - Research Associate ; Eric S Land- Research Assistant		
COI Name (Institution):	Brown, Christopher (North Carolina State University) Sederoff, Heike (North Carolina State University)		
Grant/Contract No.:	NNX10AM72G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants provide a complete and economical means for human life support for long-term space exploration and habitation. However, since the space environment is not optimal for plant growth, an understanding of how plants sense and respond to changes in their environment is of fundamental importance. The phosphoinositide (PI) pathway is highly conserved among eukaryotes and functions in the regulation of a multitude of cellular pathways. The lipid-derived second messenger inositol 1,4,5-trisphosphate (InsP3) increases in response to many different stresses. We have shown that InsP3 levels increase with gravistimulation prior to visible bending in both monocot and dicot systems. We have generated transgenic Arabidopsis plants expressing the mammalian type I inositol polyphosphate 5-phosphatase (InsP 5-ptase), an enzyme that specifically hydrolyzes InsP3 and terminates the signal. The transgenic plants have normal growth and morphology; however, they exhibit altered responses to many environmental stimuli including gravity, drought, and cold. While rapid changes in transcript levels occur in wild type Arabidopsis within 5 min of gravistimulation, the expression of several of the fastest responding genes does not change in the InsP 5-ptase roots in response to reorientation. Our hypothesis is that InsP3 is an important second messenger in the sensing and signaling of stimuli (including gravity). The plants with compromised InsP3 signaling therefore provide a valuable tool for dissecting the role of the InsP3 pathway in plant responses to the microgravity environment encountered on the International Space Station. Our primary aim is to identify the molecular changes that are specifically mediated by InsP3 in the space environment. We will compare transcript and protein profiles of wild type and transgenic plants that are grown in both microgravity and 1g in space as well as on the ground. The long term goal is to understand the molecular mechanisms plants use to sense and respond to changes in their environment. This knowledge will help design plants which are better able to withstand space flight and microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The InsP 5-ptase transgenic plants provide a good model system for understanding several issues including plant responses to gravity, plant responses to environmental stress, and fundamental aspects of PI signaling. Our previous work has implicated InsP3 as a key player in gravity sensing and signaling and the transgenic plants have attenuated InsP3 signaling and reduced gravitropism. In addition these plants have altered responses to different environmental stresses. Therefore this study will help us understand plant signaling during environmental stress, which has long term implications with improving crops to better withstand unfavorable conditions. Furthermore, the phosphoinositide pathway is highly conserved among plants, yeast, and mammals. These studies using the transgenic plants with attenuated InsP3 signaling will provide valuable insights on the regulation and downstream effectors of this pathway and could also contribute to our understanding of the role of the PI pathway in regulating human physiology.
Task Progress:	We have successfully isolated RNA from root samples from the flight experiment and carried out a comprehensive transcriptomic analysis of wild type and transgenic roots using RNA-Seq. Preliminary results show that there are significant changes between 1 g and µg conditions as well as between wild type and transgenic plants. Work is underway to compare differential gene expression in shoot samples. We anticipate that the Plant Signaling experiment will provide a comprehensive dataset that will be a valuable resource for other researchers in the field. Presentations: 1. The PI was an invited speaker at the 29th Annual meeting of American Society for Gravitational and Space Research (ASGSR), Nov 2013. Title: “Plant Signaling in Microgravity” in Section: Plants 1. 2. Student poster presentation at the 29th Annual meeting of American Society for Gravitational and Space Research (ASGSR), Nov 2013. “Quantitative analysis of Arabidopsis root growth in microgravity” K Joshi, E S Land, M Harrison-Pitaniello, C Brown, H Sederoff and IY Perera. First place winner in student poster competition (Plant Section). 3. The PI gave a guest lecture on the Plant Signaling experiment and RNA-Seq analysis in special topics course at NCSU on Deep Sequencing (BIT 815), in Feb 2014.
Bibliography Type:	Description: (Last Updated: 05/30/2025)
Abstracts for Journals and Proceedings	Land ES, Smith CM, Brown CS, Sederoff H, Perera IY. "Plant Signaling in Microgravity." 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. I.Y. Perera, invited speaker, in Section: Plants 1 session. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts, p. 70. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Joshi K, Land ES, Harrison-Pitaniello M, Brown C, Sederoff H, Perera IY. "Quantitative analysis of Arabidopsis root growth in microgravity." 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts, p. 134. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Awards	Joshi K, Land ES, Harrison-Pitaniello M, Brown C, Sederoff H, Perera IY. "K. Joshi--First place winner in student poster competition (Plant Section) for 'Quantitative analysis of Arabidopsis root growth in microgravity', 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013." Nov-2013